

HW3 Profiling Exercise Report

Barbara Klaudel

link to my repo: <https://github.com/A-Huli/bmi500/tree/dev>

a) Profiling exercise on pbmc3k, pbmc6k and pbmc10k

I run the profiling for the pbmc3k, pbmc6k and pbmc10k datasets an using scanpy_pbmc.py from the class repository with the packages from its requirements.txt installed in a Python 3.13 virtual environment. Since the environment used Python 3.13, I had to upgrade ty typing_extensions had to be upgraded and setuptools<81 installed for the packages to import. I added the profiling code directly to scanpy_pbmc.py. All runs were executed on the BMI cluster as Slurm jobs with 1 CPU core and 16 GB of memory. I profiled the pipeline at three levels:

- **Coarse-grain** - which reports the elapsed time, CPU utilization and maximum memory.
- **Instrumented** - each code section marked with `###` is timed with `time.time()`.
- **Fine-grain** - `sc.tl.rank_genes_group` run through `cProfile.run`.

b) Slurm batch scripts

Each dataset has its own batch script (the only difference in each script is the dataset name).

pbmc3k.sbatch

```
#!/bin/bash
#SBATCH -J pbmc3k
#SBATCH -N 1
#SBATCH -n 1
#SBATCH --mem=16G
#SBATCH -t 01:00:00
#SBATCH -o pbmc3k-%j.out
```

```
source bmi500/bin/activate
```

```
/usr/bin/time -v python scanpy_pbmc.py --data-dir data --data-set pbmc3k --out-dir data
--num-threads 1
```

pbmc6k.sbatch

```
#!/bin/bash
#SBATCH -J pbmc6k
#SBATCH -N 1
#SBATCH -n 1
#SBATCH --mem=16G
#SBATCH -t 01:00:00
#SBATCH -o pbmc6k-%j.out
```

```
source bmi500/bin/activate
```

```
/usr/bin/time -v python scanpy_pbmc.py --data-dir data --data-set pbmc6k --out-dir data
--num-threads 1
```

pbmc10k.sbatch

```
#!/bin/bash
#SBATCH -J pbmc10k
#SBATCH -N 1
#SBATCH -n 1
#SBATCH --mem=16G
#SBATCH -t 01:00:00
#SBATCH -o pbmc10k-%j.out
```

source bmi500/bin/activate

```
/usr/bin/time -v python scanpy_pbmc.py --data-dir data --data-set pbmc10k --out-dir data
--num-threads 1
```

c) Console outputs

The console output of each run is shown below. The coarse-grain time, CPU and memory lines and the major bottleneck from the instrumented profile are highlighted.

pbmc3k (pbmc3k-1290575.out)

```
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/scanpy/_utils/_init_.py:27: FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_csv from `anndata` is deprecated. Import
anndata.io.read_csv instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_excel from `anndata` is deprecated. Import
anndata.io.read_excel instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_hdf from `anndata` is deprecated. Import
anndata.io.read_hdf instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_loom from `anndata` is deprecated. Import
anndata.io.read_loom instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_mtx from `anndata` is deprecated. Import
anndata.io.read_mtx instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_text from `anndata` is deprecated. Import
anndata.io.read_text instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_umi_tools from `anndata` is deprecated. Import
anndata.io.read_umi_tools instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/louvain/_init_.py:54: UserWarning: pkg_resources is deprecated as an API. See
https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to
Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/scanpy/logging.py:151: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
  yield dist_name, imp.__version__
... reading from cache file cache/data-pbmc3k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 19024 genes that are detected in less than 3 cells
normalizing counts per cell
finished (0:00:00)
extracting highly variable genes
finished (0:00:00)
-> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/bkklau/bmi500/bmi500/lib/python3.13/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
```

```

... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/preprocessing/_pca.py:208: FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:00)
computing neighbors
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/neighbors/_init_.py:411: FutureWarning: The method obsm_keys is deprecated and will be removed in
the future. Use obsm instead of obsm_keys. (e.g. `k` in `adata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" in adata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished: added to `.uns['neighbors']`
    `.obsp['distances']`, distances for each pair of neighbors
    `.obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:05)
ranking genes
  finished: added to `.uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
TIME imports: 1.841 s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.8.2
pynndescent==0.6.0
using 1 threads
TIME settings: 4.922 s
TIME arguments: 0.001 s
TIME I/O read: 0.100 s
TIME filtering: 0.038 s
TIME normalize: 0.010 s
TIME highly variable genes: 0.021 s
TIME scale: 0.029 s
TIME pca: 0.115 s
TIME neighbors: 3.325 s
TIME louvain: 0.085 s
TIME umap: 5.314 s
TIME I/O write: 0.107 s
TIME rank_genes_groups: 1.268 s
Wed Sep 16 14:17:16 2026  data/pbmc3k.prof

```

42159 function calls (41000 primitive calls) in 1.258 seconds

Ordered by: cumulative time
List reduced from 1031 to 20 due to restriction <20>

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
24/1	0.000	0.000	1.258	1.258	{built-in method builtins.exec}
1	0.000	0.000	1.258	1.258	_init_.py:83(fn_compatible)
1	0.000	0.000	1.258	1.258	_rank_genes_groups.py:469(rank_genes_groups)
1	0.001	0.001	1.220	1.220	_rank_genes_groups.py:398(compute_statistics)
5	0.003	0.001	1.192	0.238	_rank_genes_groups.py:277(wilcoxon)
5	0.006	0.001	1.017	0.203	_rank_genes_groups.py:45(_ranks)
4	0.000	0.000	0.956	0.239	generic.py:10012(rank)
4	0.000	0.000	0.956	0.239	generic.py:10127(ranker)
4	0.955	0.239	0.956	0.239	algorithms.py:1067(rank)
1	0.000	0.000	0.070	0.070	_rank_genes_groups.py:174(_basic_stats)
12	0.000	0.000	0.068	0.006	_index.py:29(_getitem_)
60/59	0.000	0.000	0.067	0.001	indexing.py:1177(_getitem_)
41	0.000	0.000	0.064	0.002	indexing.py:1690(_getitem_tuple)
8	0.000	0.000	0.064	0.008	_csc.py:140(_get_arrayXSlice)
8	0.000	0.000	0.064	0.008	_compressed.py:651(_minor_index_fancy)
57	0.000	0.000	0.064	0.001	indexing.py:1720(_getitem_axis)
23	0.000	0.000	0.062	0.003	indexing.py:1206(_getbool_axis)
23	0.000	0.000	0.061	0.003	generic.py:4164(_take_with_is_copy)
23	0.000	0.000	0.061	0.003	generic.py:4049(take)
16	0.000	0.000	0.060	0.004	indexing.py:1005(_getitem_tuple_same_dim)

```

section seconds
imports 1.840509
settings 4.921953
arguments 0.000642
I/O read 0.100266
filtering 0.037756
normalize 0.009823
highly variable genes 0.020834
scale 0.028707
pca 0.115113
neighbors 3.325389
louvain 0.084749
umap 5.313554
I/O write 0.106534
rank_genes_groups 1.268286
TOTAL: 17.174 s
MAJOR BOTTLENECK: umap (5.314 s)

```

Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc3k --out-dir data --num-threads 1"

User time (seconds): 16.74

System time (seconds): 0.68

Percent of CPU this job got: 98%

Elapsed (wall clock) time (h:mm:ss or m:ss): 0:17.71

Average shared text size (kbytes): 0

Average unshared data size (kbytes): 0

Average stack size (kbytes): 0

Average total size (kbytes): 0

Maximum resident set size (kbytes): 869432

Average resident set size (kbytes): 0

Major (requiring I/O) page faults: 84

Minor (reclaiming a frame) page faults: 159640

Voluntary context switches: 10940

Involuntary context switches: 86

Swaps: 0

File system inputs: 56

File system outputs: 86656

Socket messages sent: 0

Socket messages received: 0

Signals delivered: 0

Page size (bytes): 4096

Exit status: 0

Coarse-grain: elapsed time 17.71 s, CPU utilization 98%, maximum memory 849 MB0.

Major bottleneck (instrumented): umap, 5.314 s of 17.174 s in total (30.9%).

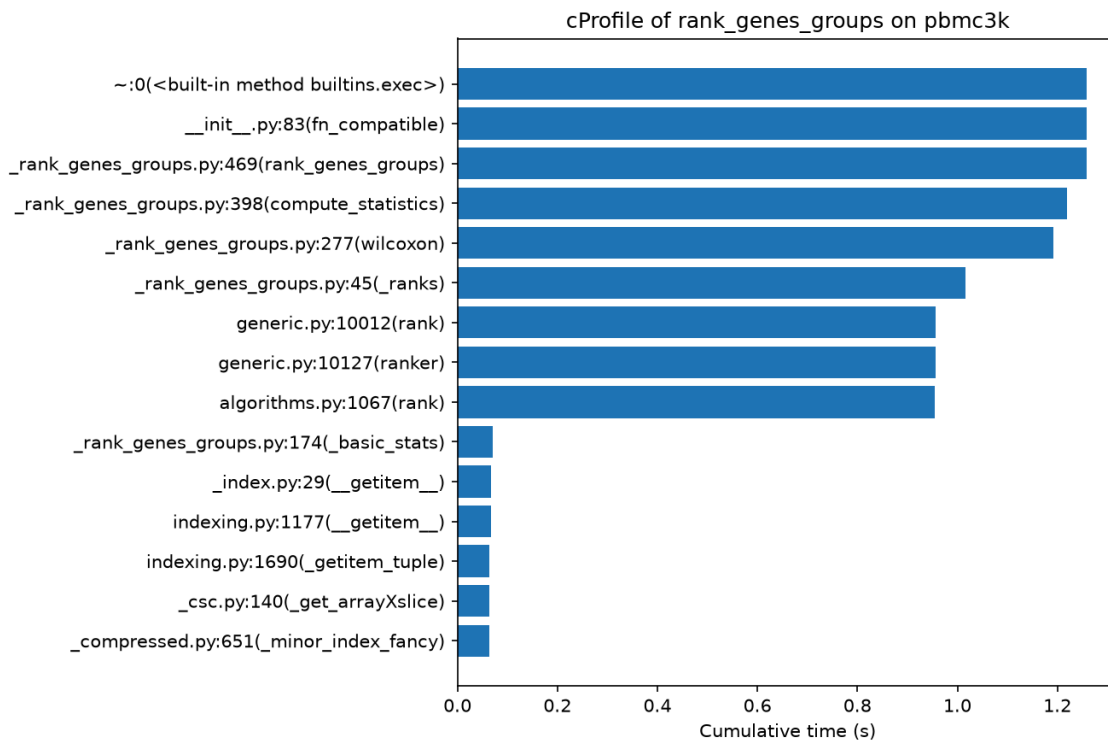


Figure 1. cProfile of rank_genes_groups on pbmc3k: top 15 functions by cumulative time.

pbmc6k (pbmc6k-1290576.out)

```
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/_utils/__init__.py:27: FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_csv from `anndata` is deprecated. Import
anndata.io.read_csv instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_excel from `anndata` is deprecated. Import
anndata.io.read_excel instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_hdf from `anndata` is deprecated. Import
anndata.io.read_hdf instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
```

```

/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_loom from `anndata` is deprecated. Import
anndata.io.read_loom instead.
    return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_mtx from `anndata` is deprecated. Import
anndata.io.read_mtx instead.
    return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_text from `anndata` is deprecated. Import
anndata.io.read_text instead.
    return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/_init_.py:70: FutureWarning: Importing read_umi_tools from `anndata` is deprecated. Import
anndata.io.read_umi_tools instead.
    return module_get_attr_redirect(attr_name, deprecated_mapping=_DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/louvain/_init_.py:54: UserWarning: pkg_resources is deprecated as an API. See
https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to
Setuptools<81.
    from pkg_resources import get_distribution, DistributionNotFound
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/logging.py:151: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
    yield dist_name, imp.__version__
... reading from cache file cache/data-pbmc6k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
normalizing counts per cell
finished (0:00:00)
extracting highly variable genes
finished (0:00:00)
--> added
    'highly_variable', boolean vector (adata.var)
    'means', float vector (adata.var)
    'dispersions', float vector (adata.var)
    'dispersions_norm', float vector (adata.var)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an AnnData. Making a copy.
    view_to_actual(adata)
... as 'zero_center=True', sparse input is densified and may lead to large memory consumption
computing PCA
    with n_comps=30
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/preprocessing/_pca.py:208: FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
    Version(ad.__version__) < Version("0.9")
    finished (0:00:00)
computing neighbors
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/neighbors/_init_.py:411: FutureWarning: The method obsm_keys is deprecated and will be removed in
the future. Use obsm instead of obsm_keys. (e.g. `k` in `adata.obsm` or `adata.obsm.keys() | {'u'}`)
    if "X_diffmap" in adata.obsm_keys():
        using 'X_pca' with n_pcs = 30
        finished: added to `uns['neighbors']`
        `obsp['distances']`, distances for each pair of neighbors
        `obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
    using the "louvain" package of Traag (2017)
    finished: found 7 clusters and added
    'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
    finished: added
    'X_umap', UMAP coordinates (adata.obsm) (0:00:10)
ranking genes
    finished: added to `uns['rank_genes_groups']`
    'names', sorted np.recarray to be indexed by group ids
    'scores', sorted np.recarray to be indexed by group ids
    'logfoldchanges', sorted np.recarray to be indexed by group ids
    'pvals', sorted np.recarray to be indexed by group ids
    'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
TIME imports: 1.840 s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.8.2
pynndescent==0.6.0
using 1 threads
TIME settings: 4.922 s
TIME arguments: 0.001 s
TIME I/O read: 0.131 s
TIME filtering: 0.057 s
TIME normalize: 0.014 s
TIME highly variable genes: 0.024 s
TIME scale: 0.057 s
TIME pca: 0.304 s
TIME neighbors: 3.391 s
TIME louvain: 0.235 s
TIME umap: 10.332 s
TIME I/O write: 0.170 s
TIME rank_genes_groups: 2.757 s
Wed Sep 16 14:17:23 2026    data/pbmc6k.prof

```

80024 function calls (78233 primitive calls) in 2.752 seconds

Ordered by: cumulative time

List reduced from 1031 to 20 due to restriction <20>

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
24/1	0.000	0.000	2.752	2.752	{built-in method builtins.exec}
1	0.000	0.000	2.752	2.752	<string>:1(<module>)
1	0.000	0.000	2.752	2.752	_init_.py:83(fn_compatible)
1	0.000	0.000	2.752	2.752	_rank_genes_groups.py:469(rank_genes_groups)
1	0.002	0.002	2.679	2.679	_rank_genes_groups.py:398(compute_statistics)
8	0.003	0.000	2.643	0.330	_rank_genes_groups.py:277(wilcoxon)

```

9 0.019 0.002 2.128 0.236 _rank_genes_groups.py:45(_ranks)
8 0.000 0.000 2.001 0.250 generic.py:10012(rank)
8 0.000 0.000 2.001 0.250 generic.py:10127(ranker)
8 1.999 0.250 1.999 0.250 algorithms.py:1067(rank)
1 0.000 0.000 0.264 0.264 _rank_genes_groups.py:174(_basic_stats)
22 0.000 0.000 0.252 0.011 _index.py:29(__getitem__)
14 0.000 0.000 0.245 0.017 _csc.py:140(_get_arrayXslice)
14 0.000 0.000 0.245 0.017 _compressed.py:651(_minor_index_fancy)
14 0.217 0.016 0.217 0.016 {built-in method scipy.sparse._sparsetools.csr_column_index2}
155/154 0.001 0.000 0.176 0.001 indexing.py:1177(__getitem__)
96 0.000 0.000 0.170 0.002 indexing.py:1690(_getitem_tuple)
152 0.000 0.000 0.169 0.001 indexing.py:1720(_getitem_axis)
63 0.000 0.000 0.160 0.003 indexing.py:1206(_getbool_axis)
56 0.000 0.000 0.158 0.003 indexing.py:1005(_getitem_tuple_same_dim)

section seconds
imports 1.840435
settings 4.921984
arguments 0.000642
I/O read 0.130663
filtering 0.056852
normalize 0.013541
highly variable genes 0.024126
scale 0.056816
pca 0.304431
neighbors 3.390595
louvain 0.234628
umap 10.331827
I/O write 0.170320
rank_genes_groups 2.757137
TOTAL: 24.234 s
MAJOR BOTTLENECK: umap (10.332 s)
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc6k --out-dir data --num-threads 1"
User time (seconds): 23.66
System time (seconds): 0.88
Percent of CPU this job got: 99%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:24.74
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 897636
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 88
Minor (reclaiming a frame) page faults: 200940
Voluntary context switches: 11571
Involuntary context switches: 97
Swaps: 0
File system inputs: 24
File system outputs: 160376
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0

```

Coarse-grain: elapsed time 24.74s, CPU utilization 99%, maximum memory 877 MB.

Major bottleneck (instrumented): umap, 10.332 s of 24.234 s in total (42.6%).

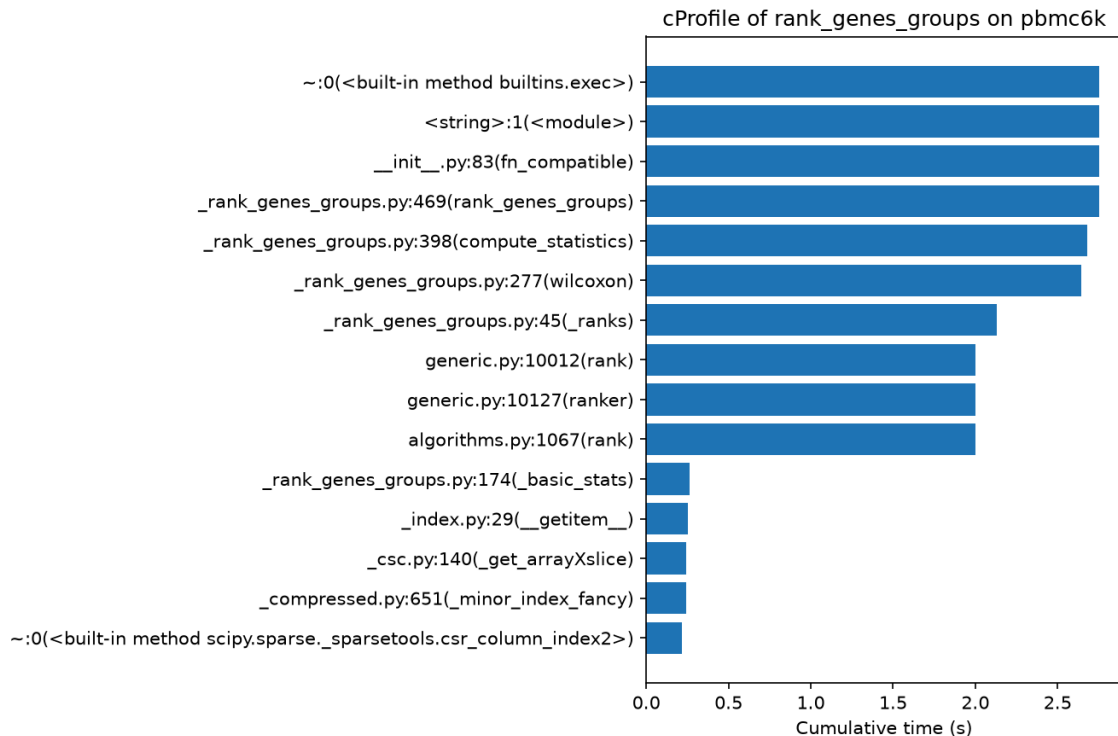


Figure 2. cProfile of rank_genes_groups on pbmc6k: top 15 functions by cumulative time.

pbmc10k (pbmc10k-1290577.out)

```
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/_utils/__init__.py:27: FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_csv from `anndata` is deprecated. Import
anndata.io.read_csv instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_excel from `anndata` is deprecated. Import
anndata.io.read_excel instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_hdf from `anndata` is deprecated. Import
anndata.io.read_hdf instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_loom from `anndata` is deprecated. Import
anndata.io.read_loom instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_mtx from `anndata` is deprecated. Import
anndata.io.read_mtx instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_text from `anndata` is deprecated. Import
anndata.io.read_text instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/anndata/__init__.py:70: FutureWarning: Importing read_umi_tools from `anndata` is deprecated. Import
anndata.io.read_umi_tools instead.
  return module_get_attr_redirect(attr_name, deprecated_mapping= _DEPRECATED)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an API. See
https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to
Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/logging.py:151: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
  yield dist_name, imp.__version__
... reading from cache file cache/data-pbmc10k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
normalizing counts per cell
finished (0:00:00)
extracting highly variable genes
finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/bklaude/bmi500/bmi500/lib/python3.13/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
```

```

... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
/home/bklaude/bmi500/lib/python3.13/site-packages/scanpy/preprocessing/_pca.py:208: FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:00)
computing neighbors
/home/bklaude/bmi500/lib/python3.13/site-packages/scanpy/neighbors/_init_.py:411: FutureWarning: The method obsm_keys is deprecated and will be removed in
the future. Use obsm instead of obsm_keys. (e.g. `k in adata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" in adata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished: added to `uns['neighbors']`
    `obsp['distances']`, distances for each pair of neighbors
    `obsp['connectivities']`, weighted adjacency matrix (0:00:21)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:09)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:10)
TIME imports: 2.031 s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.8.2
pyndescend==0.6.0
using 1 threads
TIME settings: 4.827 s
TIME arguments: 0.001 s
TIME I/O read: 0.501 s
TIME filtering: 0.287 s
TIME normalize: 0.087 s
TIME highly variable genes: 0.114 s
TIME scale: 0.128 s
TIME pca: 0.561 s
TIME neighbors: 21.598 s
TIME louvain: 0.480 s
TIME umap: 9.007 s
TIME I/O write: 0.417 s
TIME rank_genes_groups: 10.829 s
Wed Sep 16 14:17:49 2026  data/pbmc10k.prof

```

323862 function calls (318054 primitive calls) in 10.833 seconds

Ordered by: cumulative time

List reduced from 1065 to 20 due to restriction <20>

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
25/1	0.000	0.000	10.824	10.824	{built-in method builtins.exec}
1	0.000	0.000	10.824	10.824	<string>:1(<module>)
1	0.001	0.001	10.824	10.824	_init_.py:83(fn_compatible)
1	0.000	0.000	10.548	10.548	threading.py:641(wait)
1	0.003	0.003	10.531	10.531	_rank_genes_groups.py:398(compute_statistics)
17	0.011	0.001	10.432	0.614	_rank_genes_groups.py:277(wilcoxon)
25	0.048	0.002	7.041	0.282	_rank_genes_groups.py:45(_ranks)
24	0.000	0.000	6.688	0.279	generic.py:10012(rank)
24	0.001	0.000	6.688	0.279	generic.py:10127(ranker)
24	6.682	0.278	6.683	0.278	algorithms.py:1067(rank)
56	0.021	0.000	2.124	0.038	_index.py:29(_getitem_)
32	0.000	0.000	2.067	0.065	_csc.py:140(_get_arrayXslice)
32	0.001	0.000	2.067	0.065	_compressed.py:651(_minor_index_fancy)
1	0.008	0.008	1.784	1.784	_rank_genes_groups.py:174(_basic_stats)
32	1.767	0.055	1.767	0.055	{built-in method scipy.sparse._sparsetools.csr_column_index2}
856/855	0.003	0.000	0.799	0.001	indexing.py:1177(_getitem_)
469	0.002	0.000	0.774	0.002	indexing.py:1690(_getitem_tuple)
853	0.002	0.000	0.772	0.001	indexing.py:1720(_getitem_axis)
384	0.002	0.000	0.751	0.002	indexing.py:1005(_getitem_tuple_same_dim)
391	0.001	0.000	0.751	0.002	indexing.py:1206(_getbool_axis)

section	seconds
imports	2.031193
settings	4.826978
arguments	0.000578
I/O read	0.500716
filtering	0.286889
normalize	0.086696
highly variable genes	0.114004
scale	0.127751
pca	0.560756
neighbors	21.597885
louvain	0.480279
umap	9.006700
I/O write	0.417484
rank_genes_groups	10.829272
TOTAL:	50.867 s

MAJOR BOTTLENECK: neighbors (21.598 s)

Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc10k --out-dir data --num-threads 1"

User time (seconds): 48.18

System time (seconds): 3.18

Percent of CPU this job got: 99%

Elapsed (wall clock) time (h:mm:ss or m:ss): 0:51.65

Average shared text size (kbytes): 0

Average unshared data size (kbytes): 0

Average stack size (kbytes): 0

Average total size (kbytes): 0

Maximum resident set size (kbytes): 1342920

Average resident set size (kbytes): 0

Major (requiring I/O) page faults: 1973

Minor (reclaiming a frame) page faults: 692428

Voluntary context switches: 20082

Involuntary context switches: 150

Swaps: 0

File system inputs: 649200

File system outputs: 589048

Socket messages sent: 0

Socket messages received: 0

Signals delivered: 0

Page size (bytes): 4096

Exit status: 0

Coarse-grain: elapsed time 51.65 s, CPU utilization 99%, maximum memory 1,311 MB.

Major bottleneck (instrumented): neighbors, 21.598 s of 50.867 s in total (42.5%).

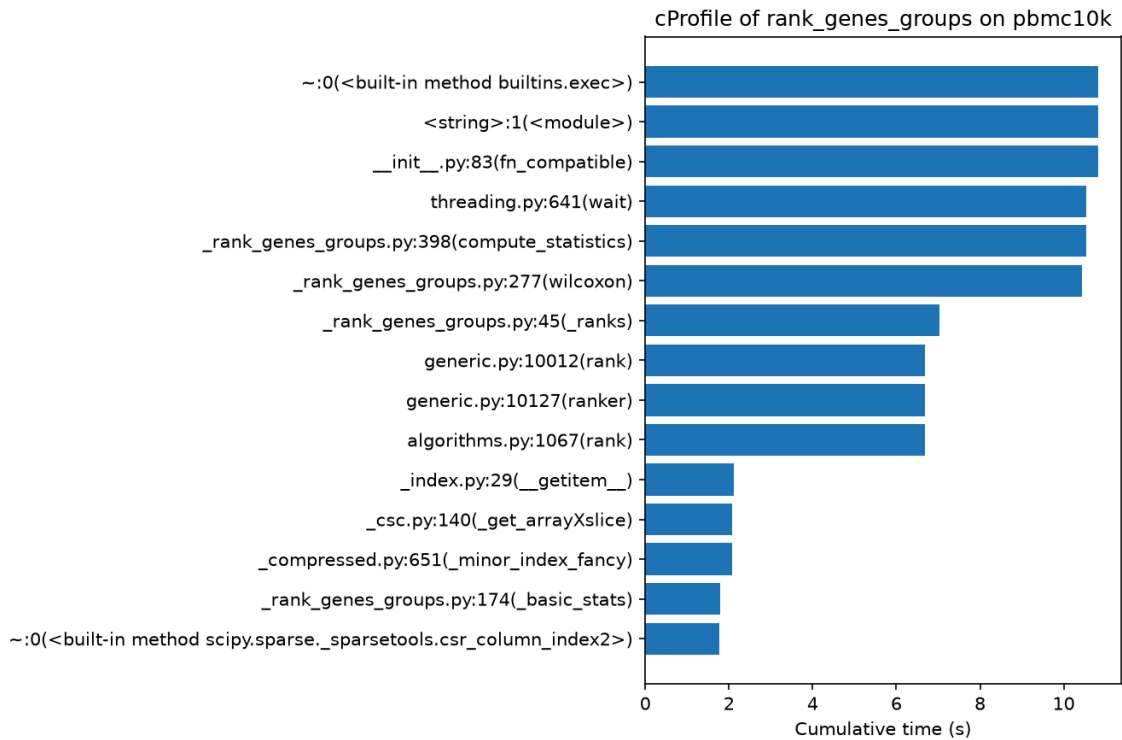


Figure 3. cProfile of rank_genes_groups on pbmc10k: top 15 functions by cumulative time.

d) Instrumented profile vs dataset size

Table 1 lists the run time of each code section, and Figure 4 shows the same data as a stacked bar chart.

Code section	pbmc3k (s)	pbmc6k (s)	pbmc10k (s)
imports	1.841	1.840	2.031
settings	4.922	4.922	4.827
arguments	0.001	0.001	0.001
I/O read	0.100	0.131	0.501
filtering	0.038	0.057	0.287
normalize	0.010	0.014	0.087
highly variable genes	0.021	0.024	0.114
scale	0.029	0.057	0.128
pca	0.115	0.304	0.561
neighbors	3.325	3.391	21.598
louvain	0.085	0.235	0.480
umap	5.314	10.332	9.007
I/O write	0.107	0.170	0.417
rank_genes_groups	1.268	2.757	10.829
TOTAL	17.174	24.234	50.867

Table 1. Instrumented profile: run time of each code section.

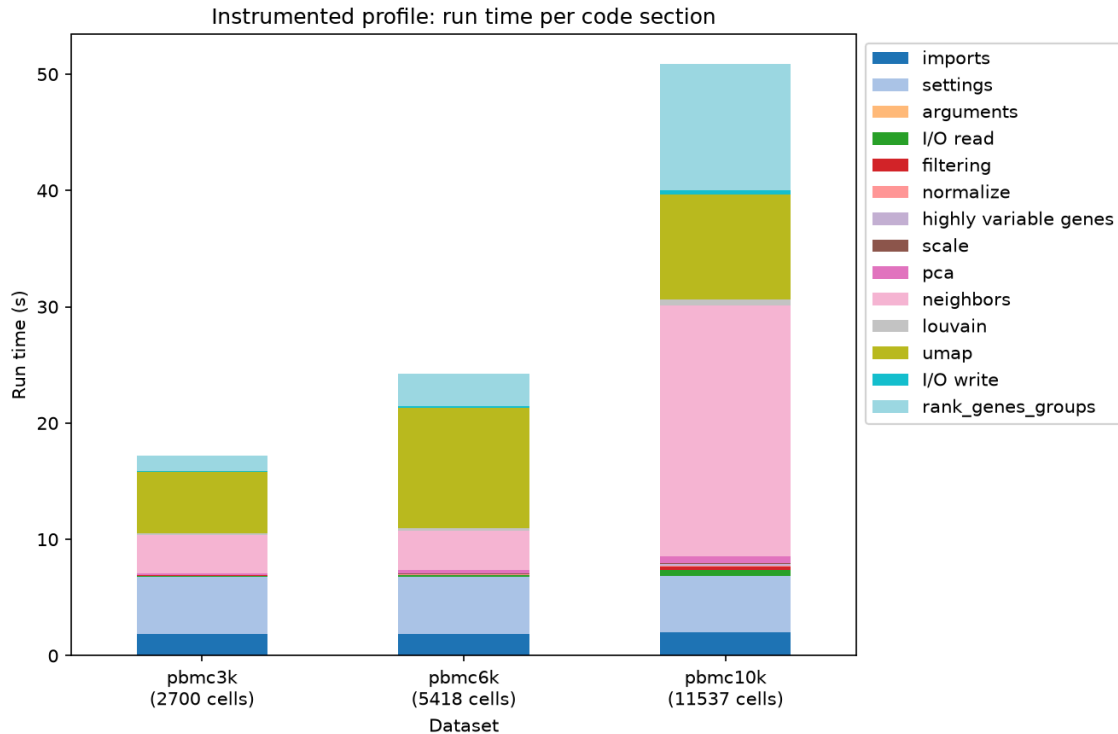


Figure 4. Run time of each code section vs dataset size.

The total run time grew from 17.2 s for pbmc3k (2,700 cells) to 24.2 s for pbmc6k (5,418 cells) and 50.9 s for pbmc10k (11,537 cells). It can be seen that the growth is not proportional to the number of cells. The fixed-cost sections, imports (1.8-2.0 s) and settings (4.8-4.9 s), take about the same time for every dataset, so they account for 39% of the pbmc3k run but only 13% of the pbmc10k run. The sections that process the data, mainly neighbors, umap and rank_genes_groups, grow much faster and account for 81% of the pbmc10k run.

e) Performance bottlenecks and expected running times at 20K cells

The main performance bottlenecks are neighbors (21.6 s, 42.5% of the pbmc10k run), rank_genes_groups (10.8 s, 21.3%) and umap (9.0 s, 17.7%). Together they take 81% of the pbmc10k run time, while reading the data and all preprocessing steps (filtering, normalization, highly variable genes and scaling) take 2.2%.

Neighbors takes about 3.4 s for pbmc3k and pbmc6k but 21.6 s for pbmc10k. Umap is slower for pbmc6k (10.3 s) than for pbmc10k (9.0 s). Rank_genes_groups grows faster than the number of cells and the number of clusters (4, 7 and 16) also increase.

To estimate the running times for 20,000 cells, I fitted $t = a \cdot n^b$ for every code section, where n is the number of cells in the three datasets, and evaluated it at $n = 20,000$ (Table 2). The exponent b shows how a section scales: $b \approx 0$ means a fixed cost and $b \approx 1$ means linear growth.

Code section	Exponent b	Expected at 20K cells (s)
imports	0.07	2.08
settings	-0.01	4.81
arguments	0.00	<0.01
I/O read	1.12	0.79
filtering	1.40	0.52
normalize	1.50	0.16
highly variable genes	1.18	0.18
scale	1.02	0.22
pca	1.09	1.09
neighbors	1.30	33.45
louvain	1.19	0.98
umap	0.36	12.49
I/O write	0.94	0.66
rank_genes_groups	1.48	22.58
TOTAL	0.75	72.67

Table 2. Expected running times at 20,000 cells.

For 20,000 cells, the fit gives a total running time of about 73 s (the estimates of the individual sections add up to 80 s). The slowest steps would be neighbors (33.5 s), rank_genes_groups (22.6 s) and umap (12.5 s).